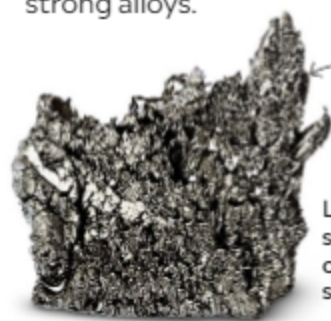


Scandium

21
Sc

Although it sits with the d-block elements, scandium isn't technically a transition element because of the way its electrons are arranged. However, it has properties similar to its neighbors. Small amounts are mixed with other metals to make strong alloys.



Scandium is shiny but tarnishes quickly in air.

Laboratory sample of pure scandium

Dmitri Mendeleev predicted the existence of scandium ten years before it was discovered in 1879.

Lightweight strength

Scandium can dramatically strengthen aluminum, even when added in tiny amounts. One such application of the scandium-aluminum alloy is in the building of fighter jets, for which high strength and low weight are crucial.



Parts of this MiG-29 jet's fuselage are composed of a scandium-aluminum alloy.

Vanadium

23
V

A typical example of a transition metal, vanadium exhibits many of the common properties of the metals in this group.

For example, it is used as a catalyst in industrial applications, as an alloy to dramatically strengthen other metals, and can also form colored compounds.

These brittle crystals are the main source of vanadium.



Vanadinite

Vanadinite

Vanadium is extracted from an ore called vanadinite. The shiny red crystals are a naturally occurring compound made of vanadium, lead, oxygen, and chlorine.

Chromium

24
Cr

Like vanadium, chromium can also form many colorful compounds—chromium is what gives the gemstone ruby its bright red color. The element was once an ingredient in pigments such as chrome yellow, but is no longer used because it is toxic.

Chrome finish

When plated on top of another metal, chromium gives machinery a polished, mirror look called the "chrome" finish. This was popular in the car industry in the 1950s and 1960s.

Chrome-plated body does not corrode easily

1967 Shelby Cobra AC Roadster 427 sports car

